

التدرب على موازنة معادلات التفاعل الكيميائي (2):

51. $\text{Li}_2\text{O} + \text{H}_2\text{O} \rightarrow \text{LiOH}$
52. $\text{CaC}_2 + \text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_2 + \text{Ca(OH)}_2$
53. $\text{Fe(OH)}_3 \rightarrow \text{Fe}_2\text{O}_3 + \text{H}_2\text{O}$
54. $\text{Pb(NO}_3)_2 \rightarrow \text{PbO} + \text{NO}_2 + \text{O}_2$
55. $\text{BaO} + \text{H}_2\text{O} \rightarrow \text{Ba(OH)}_2$
56. $\text{Ca} + \text{AlCl}_3 \rightarrow \text{CaCl}_2 + \text{Al}$
57. $\text{NH}_3 + \text{NO} \rightarrow \text{N}_2 + \text{H}_2\text{O}$
58. $\text{H}_3\text{PO}_3 \rightarrow \text{H}_3\text{PO}_4 + \text{PH}_3$
59. $\text{Fe}_2\text{O}_3 + \text{C} \rightarrow \text{CO} + \text{Fe}$
60. $\text{FeS} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2$
61. $\text{NH}_3 + \text{O}_2 \rightarrow \text{NO} + \text{H}_2\text{O}$
62. $\text{Si} + \text{S}_8 \rightarrow \text{Si}_2\text{S}_4$
63. $\text{Hg}_2\text{CO}_3 \rightarrow \text{Hg} + \text{HgO} + \text{CO}_2$
64. $\text{SiC} + \text{Cl}_2 \rightarrow \text{SiCl}_4 + \text{C}$
65. $\text{Al}_4\text{C}_3 + \text{H}_2\text{O} \rightarrow \text{CH}_4 + \text{Al(OH)}_3$
66. $\text{V}_2\text{O}_5 + \text{HCl} \rightarrow \text{VOCl}_3 + \text{H}_2\text{O}$
67. $\text{Ag}_2\text{S} + \text{KCN} \rightarrow \text{KAg(CN)}_2 + \text{K}_2\text{S}$
68. $\text{Au}_2\text{S}_3 + \text{H}_2 \rightarrow \text{Au} + \text{H}_2\text{S}$
69. $\text{ClO}_2 + \text{H}_2\text{O} \rightarrow \text{HClO}_2 + \text{HClO}_3$
70. $\text{KO}_2 + \text{CO}_2 \rightarrow \text{K}_2\text{CO}_3 + \text{O}_2$
71. $\text{MgNH}_4\text{PO}_4 \rightarrow \text{Mg}_2\text{P}_2\text{O}_7 + \text{NH}_3 + \text{H}_2\text{O}$
72. $\text{MnO}_2 + \text{HCl} \rightarrow \text{MnCl}_2 + \text{H}_2\text{O} + \text{Cl}_2$
73. $\text{Pb} + \text{Na} + \text{C}_2\text{H}_5\text{Cl} \rightarrow \text{Pb(C}_2\text{H}_5)_4 + \text{NaCl}$
74. $\text{Ca(OH)}_2 + \text{H}_3\text{PO}_4 \rightarrow \text{CaHPO}_4 + \text{H}_2\text{O}$
75. $\text{Zn} + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{Na}_2\text{Zn(OH)}_4 + \text{H}_2$
76. $\text{SrBr}_2 + (\text{NH}_4)_2\text{CO}_3 \rightarrow \text{SrCO}_3 + \text{NH}_4\text{Br}$
77. $\text{Hg(OH)}_2 + \text{H}_3\text{PO}_4 \rightarrow \text{Hg}_3(\text{PO}_4)_2 + \text{H}_2\text{O}$
78. $\text{Ca}_3(\text{PO}_4)_2 + \text{SiO}_2 + \text{C} \rightarrow \text{CaSiO}_3 + \text{P}_4 + \text{CO}$
79. $\text{I}_4\text{O}_9 \rightarrow \text{I}_2\text{O}_6 + \text{I}_2 + \text{O}_2$ (هذه المعادلة يمكن موازنتها بعدت طرق بتغيير المعاملات)
80. $\text{C}_2\text{H}_3\text{Cl} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{HCl}$
81. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \rightarrow \text{NH}_3 + \text{H}_2\text{O} + \text{Cr}_2\text{O}_3 + \text{O}_2$
82. $\text{Al} + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{NaAl(OH)}_4 + \text{H}_2$
83. $\text{NH}_4\text{Cl} + \text{Ca(OH)}_2 \rightarrow \text{CaCl}_2 + \text{NH}_3 + \text{H}_2\text{O}$
84. $\text{Al} + \text{NH}_4\text{ClO}_4 \rightarrow \text{Al}_2\text{O}_3 + \text{AlCl}_3 + \text{NO} + \text{H}_2\text{O}$
85. $\text{H}_2\text{SO}_4 + \text{NaHCO}_3 \rightarrow \text{Na}_2\text{SO}_4 + \text{CO}_2 + \text{H}_2\text{O}$
86. $\text{Ca}_{10}\text{F}_2(\text{PO}_4)_6 + \text{H}_2\text{SO}_4 \rightarrow \text{Ca(H}_2\text{PO}_4)_2 + \text{CaSO}_4 + \text{HF}$
87. $\text{Ca}_3(\text{PO}_4)_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{Ca(H}_2\text{PO}_4)_2$
88. $\text{H}_3\text{PO}_4 + (\text{NH}_4)_2\text{MoO}_4 + \text{HNO}_3 \rightarrow (\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3 + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$
89. $\text{C}_4\text{H}_{10} + \text{Cl}_2 + \text{O}_2 \rightarrow \text{CO}_2 + \text{CCl}_4 + \text{H}_2\text{O}$
90. $\text{C}_7\text{H}_{10}\text{N} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{NO}_2$
91. $\text{H}_3\text{PO}_4 + \text{HCl} \rightarrow \text{PCl}_5 + \text{H}_2\text{O}$
92. $\text{HCl} + \text{K}_2\text{CO}_3 \rightarrow \text{KCl} + \text{H}_2\text{O} + \text{CO}_2$
93. $\text{Ca(ClO}_3)_2 \rightarrow \text{CaCl}_2 + \text{O}_2$
94. $\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow \text{CO} + \text{H}_2\text{O}$
95. $\text{Xe} + \text{F}_2 \rightarrow \text{XeF}_6$
96. $\text{NH}_4\text{NO}_3 \rightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$
97. $\text{Au}_2\text{O}_3 \rightarrow \text{Au} + \text{O}_2$
98. $\text{C}_4\text{H}_{10} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
99. $\text{Fe}_3\text{O}_4 + \text{H}_2 \rightarrow \text{Fe} + \text{H}_2\text{O}$
100. $\text{O}_2 \rightarrow \text{O}_3$
101. $\text{I}_2 + \text{HNO}_3 \rightarrow \text{HIO}_3 + \text{NO}_2 + \text{H}_2$
102. $\text{C}_6\text{H}_6 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
103. $\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
104. $\text{HClO}_4 + \text{P}_4\text{O}_{10} \rightarrow \text{H}_3\text{PO}_4 + \text{Cl}_2\text{O}_7$
105. $\text{BaCl}_2 + \text{Al}_2(\text{SO}_4)_3 \rightarrow \text{BaSO}_4 + \text{AlCl}_3$
106. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \rightarrow \text{Cr}_2\text{O}_3 + \text{N}_2 + \text{H}_2\text{O}$
107. $\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$
108. $\text{Fe}_2(\text{C}_2\text{O}_4)_3 \rightarrow \text{FeC}_2\text{O}_4 + \text{CO}_2$
109. $\text{Ca}_3\text{P}_2 + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{PH}_3$
110. $\text{As} + \text{NaOH} \rightarrow \text{Na}_3\text{AsO}_3 + \text{H}_2$

51. 1,1 ---> 2
 52. 1, 2 ---> 1, 1
 53. 2 ---> 1, 3
 54. 2 ---> 2, 4, 1
 55. 1, 1 ---> 1
 56. 3, 2 ---> 3, 2
 57. 4, 6 ---> 5, 6
 58. 4 ---> 3, 1
 59. 1, 3 ---> 3, 2
 60. 4, 7 ---> 2, 4
 61. 4, 5 ---> 4, 6
 62. 4, 1 ---> 2
 63. 1 ---> 1, 1, 1
 64. 1, 2 ---> 1, 1
 65. 1, 12 ---> 3, 4
 66. 1, 6 ---> 2, 3
 67. 1, 4 ---> 2, 1
 68. 1, 3 ---> 2, 3
 69. 2, 1 ---> 1, 1
 70. 4, 2 ---> 2, 3
 71. 2 ---> 1, 2, 1
 72. 1, 4 ---> 1, 2, 1
 73. 1, 4, 4 ---> 1, 4
 74. 1, 1 ---> 1, 2
 75. 1, 2, 2 ---> 1, 1
 76. 1, 1 ---> 1, 2
 77. 3, 2 ---> 1, 6
 78. 2, 6, 10 ---> 6, 1, 10
 79. هذه المعادلة يمكن موازنتها بعدت طرق بتغيير المعاملات.
 8,9,7,9 4,1,7,15 2,2,2,3 2,1,3,6
 80. 2, 5 ---> 4, 2, 2
 81. 2 ---> 4, 2, 2, 3
 82. 2, 2, 6 ---> 2, 3
 83. 2, 1 ---> 1, 2, 2
 84. 3, 3 ---> 1, 1, 3, 6
 85. 1, 2 ---> 1, 2, 2
 86. 1, 7 ---> 3, 7, 2
 87. 1, 2 ---> 2, 1
 88. 1, 12, 21 ---> 1, 21, 12
 89. 2, 4, 11 ---> 6, 2, 10
 90. 2, 21 --> 14, 10, 2
 91. 1, 5 ---> 1, 4
 92. 2, 1 ---> 2, 1, 1
 93. 1 ---> 1, 3
 94. 1, 2 ---> 2, 3
 95. 1, 3 ---> 1
 96. 1 ---> 1, 2
 97. 2 ---> 4, 3
 98. 2, 13 ---> 8, 10
 99. 1, 4 ---> 3, 4
 100. 3 ---> 2
 101. 1, 6 ---> 2, 6, 2
 102. 2, 15 ---> 12, 6
 103. 1, 3 ---> 2, 3
 104. 12, 1 ---> 4, 6
 105. 3, 1 ---> 3, 2
 106. 1 ---> 1, 1, 4
 107. 2 ---> 1, 1, 1
 108. 1 ---> 2, 2
 109. 1, 6 ---> 3, 2
 110. 2, 6 ---> 2, 3